Linear Algebra — Lecture 5

Change of Coordinate Matrix, Dual Spaces, the Rank of a Matrix

Friedberg–Insel–Spence, *Linear Algebra* 4e, §2.5, §2.6, §3.1–§3.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §2.2–§2.3, §4.4, §4.6–§4.7

Topics. Coordinates and the coordinate mapping, the change of coordinate matrix, similar matrices, dual spaces and the transpose, and the rank of a matrix.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Coordinates, mapping	§4.4 (p. 216–221)	§2.5 (p. 110–111)
Change of coordinate matrix	§4.7 (p. 239–242)	§2.5 (p. 111–112)
Similar matrices	—	§2.5 (p. 112–115)
Linear functionals	—	§2.6 (p. 119–121)
Dual basis, transpose	—	§2.6 (p. 120–122)
The double dual	—	§2.6 (p. 121–123)
Elementary matrices	§2.2 (p. 102–108)	§3.1 (p. 148–150)
The rank of a matrix	§4.6 (p. 230–236)	§3.2 (p. 152–159)

Two questions that drive the whole lecture: (1) *After a change of basis, how do the coordinates of a vector and the matrix of an operator change?* (2) *How does the number "the dimension of the image of a matrix" behave under a change of basis, under elementary operations and under multiplication?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Coordinates and the Change of Coordinate Matrix

Lecture 3 turned an ordered basis $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ of V into a coordinate machine: every $\boldsymbol{x} \in V$ is written uniquely as $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$, and $[\boldsymbol{x}]_\beta$ is that coefficient vector. This first section does one thing: it **changes the machine** and asks how the coordinates and the matrices move with it.

Definition 1.1 (Coordinate vector; Lay §4.4, p. 216)**.** Let $\beta = \{\boldsymbol{b}_1, \dots, \boldsymbol{b}_n\}$ be an ordered basis of V. For $\boldsymbol{x} \in V$, the scalars $c_1, \dots, c_n$ in $\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ are the **coordinates** of $\boldsymbol{x}$ relative to β, the column

$$[\boldsymbol{x}]_\beta = (c_1, \dots, c_n)^\top$$

is the **coordinate vector** of $\boldsymbol{x}$, and $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is the **coordinate mapping**.

Theorem 1.1 (Unique Representation Theorem; Lay Theorem 7, §4.4, p. 216)**.** *Let β be a basis of V. Then every $\boldsymbol{x} \in V$ has exactly one expansion*

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$$

in the vectors of β.

Proof. Existence is "β spans V"; uniqueness is "β is linearly independent": if $\sum_i c_i \boldsymbol{b}_i = \sum_i d_i \boldsymbol{b}_i$, then $\sum_i (c_i - d_i)\boldsymbol{b}_i = \mathbf{0}$, so all $c_i - d_i = 0$. □

Theorem 1.2 (The coordinate mapping is an isomorphism; Lay Theorem 8, §4.4, p. 219)**.** *Let β be an ordered basis of an n-dimensional space V. Then $\boldsymbol{x} \mapsto [\boldsymbol{x}]_\beta$ is an **isomorphism** from V onto $\mathbb{F}^n$.*

Proof. Linearity: coordinates are determined by the basis, so if $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$ and $\boldsymbol{y} = \sum_i d_i \boldsymbol{b}_i$, then $\boldsymbol{x} + \boldsymbol{y} = \sum_i (c_i + d_i)\boldsymbol{b}_i$, that is, coordinates add componentwise; the same holds for scalar multiples. Injectivity: if $[\boldsymbol{x}]_\beta = \mathbf{0}$, then $\boldsymbol{x} = \sum_i 0 \cdot \boldsymbol{b}_i = \mathbf{0}$. Surjectivity: given $(c_1, \ldots, c_n)^\top$, take $\boldsymbol{x} = \sum_i c_i \boldsymbol{b}_i$. □

Definition 1.2 (Change of coordinate matrix; Friedberg §2.5, p. 110–111)**.** Let β, β' be two ordered bases of V. The matrix

$$Q = [I_V]^{\beta}_{\beta'}$$

is the **change of coordinate matrix** that changes β'-coordinates into β-coordinates; by the definition of a matrix representation, its **j-th column** is the coordinate vector $[\boldsymbol{b}'_j]_\beta$ of the j-th vector of β'.

Theorem 1.3 (How coordinates change; Friedberg Theorem 2.22, §2.5, p. 111)**.**

(a) Q is invertible;
(b) for every $\boldsymbol{v} \in V$, $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$.

Proof. (a) I_V is invertible, and "invertible if and only if its matrix representation is invertible" (Lecture 4 notes, Theorem 4.3; Friedberg Theorem 2.18) makes $Q = [I_V]^{\beta}_{\beta'}$ invertible.

(b) By "the matrix acting on coordinates gives the coordinates of the image" (Lecture 4 notes, Theorem 2.2; Friedberg Theorem 2.14),

$$[\boldsymbol{v}]_\beta = [I_V(\boldsymbol{v})]_\beta = [I_V]^{\beta}_{\beta'}[\boldsymbol{v}]_{\beta'} = Q[\boldsymbol{v}]_{\beta'}.$$ □

Remark 1.1 (Three sentences about Q)**.**

- **What the columns are**: the j-th column of Q is $[\boldsymbol{b}'_j]_\beta$, that is, "what the j-th new basis vector looks like in the old basis". So Q is never found by solving equations: expand each vector of β' in β.
- **Which side to multiply on**: $[\boldsymbol{v}]_\beta = Q[\boldsymbol{v}]_{\beta'}$ turns β'-coordinates into β-coordinates by **left** multiplication by Q; conversely $[\boldsymbol{v}]_{\beta'} = Q^{-1}[\boldsymbol{v}]_\beta$. To remember the direction, read $Q = [I_V]^{\beta}_{\beta'}$: subscript in, superscript out. The same sense of direction governs the change of basis of an operator: $[T]_{\beta'}$ needs both ends in the new basis, hence Q on the left and Q^{-1} on the right, that is $Q^{-1}[T]_\beta Q$.
- **Two special cases**: if $\beta' = \beta$ then $Q = I_n$; if $V = \mathbb{F}^n$ and β is the standard basis, the columns of Q are simply the vectors of β' themselves, because coordinates in the standard basis are the entries.

Example 1.1 (Computing Q; Friedberg Example 1, §2.5, p. 112)**.** In $\mathbb{R}^2$ take $\beta = \{(1,1), (1,-1)\}$ and $\beta' = \{(2,4), (3,1)\}$. Find the matrix Q that changes β'-coordinates into β-coordinates, and compute $[(2,4)]_\beta$.

Approach. The columns of Q are the coordinate vectors of the vectors of β' in the basis β, so each of $(2,4)$ and $(3,1)$ has to be written as a combination of $(1,1)$ and $(1,-1)$; these are two small linear systems, and with these numbers the sum and the difference give the answer at once.

From

$$(2,4) = 3(1,1) - 1(1,-1), \qquad (3,1) = 2(1,1) + 1(1,-1)$$

we get

$$Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}, \qquad [(2,4)]_\beta = Q \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

The second identity is checked directly: $3(1,1) - 1(1,-1) = (2,4)$.

Definition 1.3 (Linear operator; Friedberg §2.5, p. 111)**.** Let V be a vector space over $\mathbb{F}$. A linear transformation T whose domain and codomain coincide, that is $T : V \to V$, is a **linear operator** on V; the set of all such T is written $\mathcal{L}(V)$. Two operators can be composed; the distributive, associative and identity laws for composition are exactly the four laws of Lecture 4 notes, Theorem 1.2 — in the language of structure, $\mathcal{L}(V)$ is a (non-commutative) algebra.

Theorem 1.4 (Changing the basis of an operator; Friedberg Theorem 2.23, §2.5, p. 112)**.** *Let T be a linear operator on a finite-dimensional space V, let β, β' be ordered bases of V, and let Q change β'-coordinates into β-coordinates. Then*

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q.$$

Proof. Write $I = I_V$. Since $T = IT = TI$, the matrix of a composition (Lecture 4 notes, Theorem 2.3; Friedberg Theorem 2.11) gives

$$Q[T]_{\beta'} = [I]^\beta_{\beta'}[T]^{\beta'}_{\beta'} = [IT]^\beta_{\beta'} = [TI]^\beta_{\beta'} = [T]^\beta_\beta [I]^\beta_{\beta'} = [T]_\beta Q,$$

and multiplying on the left by Q^{-1} finishes the proof. □

Proposition 1.5 (The matrix version; Friedberg §2.5, p. 115, Corollary)**.** *Let $A \in M_{n\times n}(\mathbb{F})$, let γ be an ordered basis of $\mathbb{F}^n$, and let Q be the matrix whose j-th column is the j-th vector of γ. Then*

$$[L_A]_\gamma = Q^{-1}AQ.$$

Proof. Take $T = L_A$ and β the standard basis in Theorem 1.4: then $[T]_\beta = A$, and $Q = [I]^\beta_{\beta'}$ has the vectors of γ as its columns (the third special case in Remark 1.1), so $[L_A]_\gamma = Q^{-1}AQ$. □

Definition 1.4 (Similar matrices; Friedberg §2.5, p. 115)**.** Let $A, B \in M_{n\times n}(\mathbb{F})$. If $B = Q^{-1}AQ$ for some invertible Q, then B is **similar** to A. By Theorem 1.4, the matrices of one operator in two bases are always similar.

Remark 1.2 (Similarity is an equivalence relation)**.** $A = IAI^{-1}$ gives reflexivity; $B = Q^{-1}AQ$ implies $A = QBQ^{-1}$, symmetry; and if $B = Q^{-1}AQ$ and $C = P^{-1}BP$, then $C = (QP)^{-1}A(QP)$, transitivity. So similarity classifies operators: the matrices in one class are the same operator in different clothes.

Example 1.2 (Changing the basis in practice; Friedberg Example 2, §2.5, p. 113)**.** Let T be the operator on $\mathbb{R}^2$ defined by $T(a,b) = (3a - b,\ a + 3b)$, with β, β' as in Example 1.1. Compute $[T]_{\beta'}$ and check it against the images of the basis vectors.

Approach. Take both roads. One is $Q^{-1}[T]_\beta Q$ from Theorem 1.4; the other goes back to the definition and expands $T(\boldsymbol{b}'_j)$ in β', to see whether the j-th column comes out the same. The first road is fast; the second is the only way to be sure the matrix is right.

In the standard basis $[T]_\beta = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$ (the first column is $T(1,0) = (3,1)$, the second is $T(0,1) = (-1,3)$), and Example 1.1 gives $Q = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$ with $Q^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$. Hence

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q = \frac{1}{5}\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}\begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ -2 & 2 \end{bmatrix}.$$

Check: $T(3,1) = (8,6) = 1 \cdot (2,4) + 2 \cdot (3,1)$, whose coefficients $(1,2)^\top$ are the second column (the second vector of β' is $(3,1)$); and $T(2,4) = (2,14) = -2 \cdot (2,4) + 2 \cdot (3,1)$, whose coefficients $(-2,2)^\top$ are the first column. Both columns match.

2 Dual Spaces

Push the idea of a matrix representation to its limit and ask: **can all linear transformations on V be packed into one vector space?** Yes — into V^*. Duality is "measuring vectors with linear functionals", and the rank computations of the next section use exactly this measuring tool.

Definition 2.1 (Linear functional; Friedberg §2.6, p. 119)**.** Let V be a vector space over $\mathbb{F}$. A linear map $f : V \to \mathbb{F}$ (whose codomain is the one-dimensional space $\mathbb{F}$) is a **linear functional** on V.

Example 2.1 (Three linear functionals to remember; Friedberg Examples 1–3, §2.6, p. 119)**.**

- **Integral type**: on $V = C[0, 2\pi]$ fix g and set $h(\boldsymbol{x}) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$. With $g = \sin nt$ or $\cos nt$, $h(\boldsymbol{x})$ is the n-th Fourier coefficient.
- **Trace type**: on $V = M_{n\times n}(\mathbb{F})$, $f(A) = \operatorname{tr}(A)$.
- **Coordinate type**: fix an ordered basis $\beta = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ of V and let $f_i(\boldsymbol{x}) = a_i$, where $[\boldsymbol{x}]_\beta = (a_1, \ldots, a_n)^\top$. The map f_i is the i-th **coordinate function**; the point is that $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Definition 2.2 (Dual space; Friedberg §2.6, p. 119)**.** The space $\mathcal{L}(V, \mathbb{F})$ of all linear functionals on V is the **dual space** of V, written V^*. If $\dim V = n$, then $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$ (Lecture 4 notes, Proposition 4.8) gives

$$\dim V^* = \dim V.$$

Theorem 2.1 (The dual basis; Friedberg Theorem 2.24, §2.6, p. 120)**.** *Let V be n-dimensional with ordered basis $\beta = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$, let f_i be the i-th coordinate function and $\beta^* = \{f_1, \ldots, f_n\}$. Then β^* is an ordered basis of V^*, and for every $f \in V^*$,*

$$f = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\, f_i.$$

Proof. First show that $g = \sum_i f(\boldsymbol{x}_i) f_i$ equals f: for each basis vector $\boldsymbol{x}_j$,

$$g(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i) f_i(\boldsymbol{x}_j) = \sum_{i=1}^{n} f(\boldsymbol{x}_i)\delta_{ij} = f(\boldsymbol{x}_j),$$

and two linear functionals that agree on a basis are equal (Lecture 3 notes, Theorem 3.1). So β^* spans V^*; since $|\beta^*| = n = \dim V^*$, it is a basis. □

Definition 2.3 (Dual basis; Friedberg §2.6, p. 120)**.** The ordered basis $\beta^* = \{f_1, \ldots, f_n\}$ of Theorem 2.1 is the **dual basis** of β; its defining property is $f_i(\boldsymbol{x}_j) = \delta_{ij}$.

Example 2.2 (Computing a dual basis; Friedberg Example 4, §2.6, p. 120)**.** Let $\beta = \{(2,1), (3,1)\}$ be a basis of $\mathbb{R}^2$ and find the dual basis f_1, f_2.

Approach. The conditions $f_i(\boldsymbol{b}_j) = \delta_{ij}$ are four equations whose unknowns are the values of f_i on the standard basis. Writing $\boldsymbol{b}_j$ in terms of $\boldsymbol{e}_1, \boldsymbol{e}_2$ turns them into familiar linear systems; alternatively, read the answer off directly: f_1 must take the value 1 at $(2,1)$ and 0 at $(3,1)$.

For f_1, write $f_1(x, y) = ax + by$; then

$$2a + b = 1, \qquad 3a + b = 0,$$

so $a = -1$, $b = 3$, i.e. $f_1(x, y) = -x + 3y$; similarly $f_2(x, y) = x - 2y$. Check: $f_1(2, 1) = 1$, $f_1(3, 1) = 0$, $f_2(2, 1) = 0$, $f_2(3, 1) = 1$.

Theorem 2.2 (Transpose; Friedberg Theorem 2.25, §2.6, p. 121)**.** *Let V, W be finite-dimensional with bases β, γ, and let $T : V \to W$ be linear. Define $T^t : W^* \to V^*$ by*

$$T^t(g) = g \circ T, \qquad \textit{that is,} \qquad T^t(g)(\boldsymbol{x}) = g(T(\boldsymbol{x})).$$

Then T^t is linear and

$$[T^t]_{\gamma^*}^{\beta^*} = \big([T]_\beta^\gamma\big)^\top.$$

Proof. The linearity of T^t follows from the compatibility of composition with addition and scalar multiplication. Let $\beta = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$, $\gamma = \{\boldsymbol{y}_1, \ldots, \boldsymbol{y}_m\}$ and $A = [T]_\beta^\gamma$, so that $T(\boldsymbol{x}_i) = \sum_k A_{ki}\boldsymbol{y}_k$. Expanding $T^t(g_j)$ in β^* by Theorem 2.1, its i-th coefficient is

$$(T^t(g_j))(\boldsymbol{x}_i) = g_j(T(\boldsymbol{x}_i)) = \sum_k A_{ki} g_j(\boldsymbol{y}_k) = A_{ji},$$

because $g_j(\boldsymbol{y}_k) = \delta_{jk}$; this is the entry in row i, column j of $A^\top$. □

Remark 2.1 (How to remember it)**.** Theorem 2.2 says: **the matrix of the transpose of a transformation is the transpose of its matrix**, with each side taken in the corresponding dual bases. Its use is to translate "left or right multiplication" into "transpose": the map $g \mapsto g \circ T$ and the map $A \mapsto A^\top$ are the same thing.

Example 2.3 (A transpose computed; Friedberg Example 5, §2.6, p. 122)**.** Let $T : P_1(\mathbb{R}) \to \mathbb{R}^2$ be $T(p(x)) = (p(0), p(2))$ with β, γ the standard bases. Then $[T]_\beta^\gamma = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}$ and

$$[T^t]_{\gamma^*}^{\beta^*} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \big([T]_\beta^\gamma\big)^\top.$$

The check: expand $T^t(g_1)$ in $\beta^* = \{f_1, f_2\}$; its coefficients are determined by $T^t(g_1)(1)$ and $T^t(g_1)(x)$, and they agree with the columns of the matrix.

Definition 2.4 (The element $\hat{\boldsymbol{x}}$ of the double dual; Friedberg §2.6, p. 121)**.** Let V be a vector space and $\boldsymbol{x} \in V$. Define $\hat{\boldsymbol{x}} : V^* \to \mathbb{F}$ by

$$\hat{\boldsymbol{x}}(f) = f(\boldsymbol{x}) \qquad (\text{for every } f \in V^*),$$

that is: hold $\boldsymbol{x}$ fixed and let the input be a functional — $\hat{\boldsymbol{x}}$ reads f and returns the number $f(\boldsymbol{x})$. It is linear in f, so $\hat{\boldsymbol{x}} \in V^{**}$.

Lemma 2.3 (The lemma behind the double dual; Friedberg §2.6, p. 122)**.** *Let V be finite-dimensional and $\boldsymbol{x} \in V$. If $\hat{\boldsymbol{x}}(f) = 0$ for every $f \in V^*$, then $\boldsymbol{x} = \boldsymbol{0}$.*

Proof. If $\boldsymbol{x} \neq \boldsymbol{0}$, extend it to an ordered basis $\beta = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ with $\boldsymbol{x}_1 = \boldsymbol{x}$ and take f_1 from the dual basis; then $\hat{\boldsymbol{x}}(f_1) = f_1(\boldsymbol{x}) = 1 \neq 0$. □

Theorem 2.4 (The double dual; Friedberg Theorem 2.26, §2.6, p. 122)**.** *Let V be finite-dimensional and define $\psi : V \to V^{**}$ by $\psi(\boldsymbol{x}) = \hat{\boldsymbol{x}}$. Then ψ is an isomorphism.*

Proof. Linearity: $f(\boldsymbol{x} + \boldsymbol{y}) = f(\boldsymbol{x}) + f(\boldsymbol{y})$ for every f, hence $\widehat{\boldsymbol{x} + \boldsymbol{y}} = \hat{\boldsymbol{x}} + \hat{\boldsymbol{y}}$, and similarly for scalar multiples. Injectivity: Lemma 2.3. So ψ is injective, and $\dim V^{**} = \dim V^* = \dim V$ makes it surjective as well. □

Proposition 2.5 (The converse question for dual bases; Friedberg §2.6, p. 123, Corollary)**.** *Let V be finite-dimensional. Then every ordered basis of V^* is the dual basis of some basis of V.*

Proof. Let $\{f_1, \dots, f_n\}$ be an ordered basis of V^* and let $\{F_1, \dots, F_n\}$ be its dual basis in $V^{**} = (V^*)^*$, so that $F_i(f_j) = \delta_{ij}$. The map $\psi : V \to V^{**}$ of Theorem 2.4 is onto, so there are $\boldsymbol{x}_1, \dots, \boldsymbol{x}_n \in V$ with $\hat{\boldsymbol{x}}_i = F_i$. Then for all i, j,

$$f_j(\boldsymbol{x}_i) = \hat{\boldsymbol{x}}_i(f_j) = F_i(f_j) = \delta_{ij},$$

and ψ^{-1} carries a basis to a basis, so $\{\boldsymbol{x}_1, \dots, \boldsymbol{x}_n\}$ is an ordered basis of V; hence $\{f_1, \dots, f_n\}$ is its dual basis. □

Remark 2.2 (Natural and not natural)**.** The assignment $\beta \mapsto \beta^*$ depends on β: change the basis and the whole dual basis changes (hence "V is isomorphic to V^*" is true, but no isomorphism is independent of a basis). The map ψ of Theorem 2.4, by contrast, **uses no basis at all** — it is the "natural" isomorphism, which is why we identify V with V^{**} but never with V^*.

3 The Rank of a Matrix

The last part of the lecture is **one number**: $\operatorname{rank}(A)$. It is "how many dimensions the image of A has", it is "how many columns of A are really independent", and it is "how many pivots a row reduction leaves". The task of this section is to prove that these three sentences say the same thing, and to see how the number behaves under a change of basis and under multiplication.

Definition 3.1 (Elementary operations and elementary matrices; Friedberg §3.1, p. 148–150)**.** An **elementary row operation** on a matrix is one of: interchanging two rows; multiplying a row by a nonzero scalar; adding a multiple of one row to another. Column operations are defined the same way. An $m \times m$ matrix obtained from I_m by one elementary row operation is an **elementary matrix**.

Theorem 3.1 (Elementary matrices realise elementary operations; Friedberg Theorem 3.1, §3.1, p. 149)**.** *Let $A \in M_{m\times n}(\mathbb{F})$ and suppose B is obtained from A by one elementary row (column) operation. Then there is an $m \times m$ ($n \times n$) elementary matrix E with*

$$B = EA \qquad (\textit{respectively } B = AE),$$

where E is obtained from I_m (I_n) by the same operation.

Theorem 3.2 (Elementary matrices are invertible; Friedberg Theorem 3.2, §3.1, p. 150)**.** *Every elementary matrix is invertible, and its inverse is an elementary matrix of the same type.*

Proof. Let E come from I_n by one elementary row operation. Undo that step (interchange again; multiply by $1/c$; subtract the same multiple), and E goes back to I_n by an operation of the same type. By Theorem 3.1 there is an elementary matrix E' of that type with $E'E = I_n$, so E is invertible and $E^{-1} = E'$. □

Definition 3.2 (Rank; Friedberg §3.2, p. 152; Lay §4.6, p. 233)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and let $L_A : \mathbb{F}^n \to \mathbb{F}^m$ be its left-multiplication map. Define

$$\operatorname{rank}(A) = \dim \mathcal{R}(L_A) = \dim \operatorname{Col}(A),$$

the dimension of the column space of A (the span of its columns). Then the dimension theorem (Lecture 3 notes) gives $\operatorname{rank}(A) + \operatorname{nullity}(L_A) = n$, that is $\dim \operatorname{Col}(A) + \dim \operatorname{Nul}(A) = n$.

Theorem 3.3 (Rank does not depend on the representation; Friedberg Theorem 3.3, §3.2, p. 152)**.** *Let $T : V \to W$ be linear and let β, γ be ordered bases of V, W. Then*

$$\operatorname{rank}(T) = \operatorname{rank}\big([T]_\beta^\gamma\big).$$

Proof. By Lecture 4 notes, Theorem 2.2, the maps $\boldsymbol{v} \mapsto [T(\boldsymbol{v})]_\gamma$ and $[\boldsymbol{v}]_\beta \mapsto [T]_\beta^\gamma [\boldsymbol{v}]_\beta$ are the same assignment, and the coordinate mappings are isomorphisms (Theorem 1.2), so the two images have the same dimension. □

Theorem 3.4 (Multiplying by invertible matrices preserves the rank; Friedberg Theorem 3.4, §3.2, p. 153)**.** *Let A be $m \times n$ and let P, Q be invertible of sizes $m \times m$ and $n \times n$. Then*

$$\operatorname{rank}(AQ) = \operatorname{rank}(PA) = \operatorname{rank}(PAQ) = \operatorname{rank}(A).$$

In particular (with P, Q products of elementary matrices): ***elementary row and column operations preserve the rank.***

Proof. L_Q is onto (as Q is invertible), so $\mathcal{R}(L_{AQ}) = L_A(L_Q(\mathbb{F}^n)) = L_A(\mathbb{F}^n) = \mathcal{R}(L_A)$, and taking dimensions gives $\operatorname{rank}(AQ) = \operatorname{rank}(A)$. Also $\operatorname{rank}(PA) = \operatorname{rank}((PA)^\top) = \operatorname{rank}(A^\top P^\top)$ with $P^\top$ invertible, so the result just proved applies (using $\operatorname{rank}(A) = \operatorname{rank}(A^\top)$, see Proposition 3.6). The two identities together give $\operatorname{rank}(PAQ) = \operatorname{rank}(A)$. □

Theorem 3.5 (Rank is the maximal number of independent columns; Friedberg Theorem 3.5, §3.2, p. 153)**.** *The rank of a matrix equals the maximal number of linearly independent columns, that is, the dimension of the subspace spanned by its columns.*

Proof. $\operatorname{rank}(A) = \dim \mathcal{R}(L_A)$, and $\mathcal{R}(L_A) = \operatorname{Span}\{A\boldsymbol{e}_1, \ldots, A\boldsymbol{e}_n\} = \operatorname{Span}\{\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n\}$ (Lecture 4 notes, Theorem 2.1(b)), which is exactly the column space. □

Proposition 3.6 (Row rank equals column rank; Friedberg §3.2, p. 158, Corollary 2)**.** *For every $m \times n$ matrix A,*

$$\operatorname{rank}(A^\top) = \operatorname{rank}(A),$$

and $\operatorname{rank}(A)$ *is also the maximal number of linearly independent rows; the row space and the column space have the same dimension.*

Theorem 3.7 (The normal form for a given rank; Friedberg Theorem 3.6, §3.2, p. 155)**.** *Let A be $m \times n$ of rank r. Then $r \le m$, $r \le n$, and finitely many elementary row and column operations transform A into*

$$D = \begin{bmatrix} I_r & O_1 \\ O_2 & O_3 \end{bmatrix},$$

where O_1, O_2, O_3 are zero matrices (that is $D_{ii} = 1$ for $i \le r$ and all other entries 0).

Proposition 3.8 (Two consequences of the normal form; Friedberg §3.2, p. 158–159)**.**

(a) A of rank r can be written as $D = BAC$ with B, C invertible (Corollary 1);
(b) every invertible matrix is a product of elementary matrices (Corollary 3).

Proof. (a) Every elementary operation is a left or right multiplication by an elementary matrix (Theorem 3.1); multiply those matrices in order to get B, C, which are invertible by Theorem 3.2 and by "a product of invertible matrices is invertible".

(b) If A is invertible then $\operatorname{rank}(A) = n$, so the normal form is $D = I_n$; by (a), $I_n = BAC$, hence $A = B^{-1}C^{-1}$, and B^{-1}, C^{-1} are products of elementary matrices. □

Theorem 3.9 (The Rank Theorem; Lay Theorem 14, §4.6, p. 233)**.** *Let A be $m \times n$. Then the column space and the row space have the same dimension, this common dimension is* $\operatorname{rank}(A)$*, it equals the number of pivot positions of A, and*

$$\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n.$$

Proof. The identity $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$ is the dimension theorem applied to $L_A : \mathbb{F}^n \to \mathbb{F}^m$ (Lecture 3 notes). For "$\operatorname{rank}(A)$ = the number of pivots": row reduction preserves the rank (Theorem 3.4), while in echelon form the pivot columns are linearly independent and every other column is a combination of them, so the rank is the number of pivots (Theorem 3.5). Row rank equals column rank is Proposition 3.6. □

Theorem 3.10 (Monotonicity of the rank; Friedberg Theorem 3.7, §3.2, p. 159)**.** *Let $T : V \to W$, $U : W \to Z$ be linear and let A, B be matrices for which AB is defined. Then*

$$\operatorname{rank}(UT) \le \operatorname{rank}(U), \qquad \operatorname{rank}(UT) \le \operatorname{rank}(T),$$

$$\operatorname{rank}(AB) \le \operatorname{rank}(A), \qquad \operatorname{rank}(AB) \le \operatorname{rank}(B).$$

Proof. Because

$$\mathcal{R}(UT) = U\big(\mathcal{R}(T)\big) \subseteq \mathcal{R}(U),$$

taking dimensions gives the first inequality. (b): put $A' = [U]_\gamma^\beta$ and $B' = [T]_\alpha^\beta$; by Lecture 4 notes, Theorem 2.3, $[UT]_\alpha^\gamma = A'B'$, hence by Theorem 3.3

$$\operatorname{rank}(UT) = \operatorname{rank}(A'B') \le \operatorname{rank}(B') = \operatorname{rank}(T).$$

(c)(d) are the matrix versions of (a)(b); for (d) use the transpose, $\operatorname{rank}(AB) = \operatorname{rank}(B^\top A^\top) \le \operatorname{rank}(B^\top) = \operatorname{rank}(B)$. □

Remark 3.1 (The Invertible Matrix Theorem in rank language; Lay §4.6, p. 235)**.** For an $n \times n$ matrix A, each of the following is equivalent to "A is invertible": the columns of A form a basis of $\mathbb{R}^n$; $\operatorname{Col}(A) = \mathbb{R}^n$; $\dim \operatorname{Col}(A) = n$; $\operatorname{rank}(A) = n$; $\operatorname{Nul}(A) = \{\mathbf{0}\}$; $\dim \operatorname{Nul}(A) = 0$. They continue the Invertible Matrix Theorem of Lecture 4, merely replacing the language of "$A\boldsymbol{x} = \boldsymbol{b}$" by the language of rank.

4 A Worked Example

Example 4.1 (Change of basis, dual basis and rank in one pass; written for this lecture; the methods are those of Friedberg §2.5, §2.6, §3.2 and Lay §4.6, §4.7)**.** Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 0 \end{bmatrix},$$

let $\beta' = \{(1,0,0), (1,1,0), (1,1,1)\}$ be an ordered basis of $\mathbb{R}^3$, and let β be the standard basis. (1) Write the change of coordinate matrix $Q = [I]_{\beta'}^{\beta}$ and find Q^{-1}; (2) compute $[T]_{\beta'}$ as $Q^{-1}AQ$ and check it against the images of the $\boldsymbol{b}'_j$; (3) find $\operatorname{rank}(A)$ and $\dim \operatorname{Nul}(A)$, and explain why $\operatorname{rank}([T]_{\beta'}) = \operatorname{rank}(A)$; (4) find the dual basis f_1, f_2, f_3 of β' and verify $f_i(\boldsymbol{b}'_j) = \delta_{ij}$; (5) let B be the elementary matrix obtained by interchanging rows 1 and 2 of I_3, and verify that $\operatorname{rank}(BA) = \operatorname{rank}(A)$.

Approach. Each part matches one theorem. (1) is "the columns of Q are the new basis vectors in the old basis"; here the old basis is the standard one, so the columns of Q are the three vectors themselves. (2) is Theorem 1.4. (3) is "rank = number of pivots" plus Theorem 3.3. (4) is "the

dual basis is fixed by $f_i(\boldsymbol{b}'_j) = \delta_{ij}$", read off from the formula that solves for β'-coordinates. (5) is Theorem 3.4: elementary matrices are invertible, so left multiplication by one cannot change the rank. The skeleton of the whole example is one sentence: *a change of basis changes the look of the matrix, never its rank.*

(1) The change of coordinate matrix. Since β is the standard basis, the columns of Q are the three vectors of β':

$$Q = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad Q^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(Q is upper triangular with all diagonal entries 1, so its inverse is read off immediately: $\boldsymbol{e}_1 = \boldsymbol{b}'_1$, $\boldsymbol{e}_2 = \boldsymbol{b}'_2 - \boldsymbol{b}'_1$, $\boldsymbol{e}_3 = \boldsymbol{b}'_3 - \boldsymbol{b}'_2$, and these coefficients, taken as columns, give Q^{-1}.)

(2) The matrix in the new basis. First

$$AQ = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 0 & 1 & 1 \end{bmatrix},$$

then multiply on the left by Q^{-1} (subtract the next row from each row):

$$[T]_{\beta'} = Q^{-1}AQ = \begin{bmatrix} -1 & -3 & -4 \\ 2 & 5 & 7 \\ 0 & 1 & 1 \end{bmatrix}.$$

Check: $T(\boldsymbol{b}'_1) = A(1,0,0)^\top = (1,2,0)^\top = -1\cdot\boldsymbol{b}'_1 + 2\cdot\boldsymbol{b}'_2$, whose coefficients $(-1,2,0)^\top$ are the first column; $T(\boldsymbol{b}'_2) = (3,6,1)^\top = -3\boldsymbol{b}'_1 + 5\boldsymbol{b}'_2 + 1\boldsymbol{b}'_3$, whose coefficients $(-3,5,1)^\top$ are the second column; and $T(\boldsymbol{b}'_3) = (4,8,1)^\top = -4\boldsymbol{b}'_1 + 7\boldsymbol{b}'_2 + \boldsymbol{b}'_3$, whose coefficients $(-4,7,1)^\top$ are the third column. All three match.

(3) Rank and nullity. Row-reduce A: subtract twice the first row from the second,

$$A \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so there are two pivots and $\operatorname{rank}(A) = 2$; the dimension theorem then gives $\dim \operatorname{Nul}(A) = 3-2 = 1$. Indeed, solving $A\boldsymbol{x} = \boldsymbol{0}$ gives $x_2 = 0$ and $x_1 + x_3 = 0$, that is

$$\operatorname{Nul}(A) = \operatorname{Span}\{(-1,0,1)^\top\}.$$

For $[T]_{\beta'}$ the same thing happens: subtracting rows produces a zero row (the third row is a combination of the first two), so $\operatorname{rank}([T]_{\beta'}) = 2 = \operatorname{rank}(A)$, in agreement with Theorem 3.3.

(4) The dual basis. Write $\boldsymbol{x} = (x_1, x_2, x_3)$ in the basis β': $\boldsymbol{x} = c_1\boldsymbol{b}'_1 + c_2\boldsymbol{b}'_2 + c_3\boldsymbol{b}'_3$, and comparing entries gives $c_3 = x_3$, $c_2 = x_2 - x_3$, $c_1 = x_1 - x_2$. The coordinate functions are the dual basis:

$$f_1(\boldsymbol{x}) = x_1 - x_2, \qquad f_2(\boldsymbol{x}) = x_2 - x_3, \qquad f_3(\boldsymbol{x}) = x_3.$$

Verification: $f_1(\boldsymbol{b}'_1) = 1$, $f_1(\boldsymbol{b}'_2) = 0$, $f_1(\boldsymbol{b}'_3) = 0$; $f_2(\boldsymbol{b}'_1) = 0$, $f_2(\boldsymbol{b}'_2) = 1$, $f_2(\boldsymbol{b}'_3) = 0$; $f_3(\boldsymbol{b}'_1) = f_3(\boldsymbol{b}'_2) = 0$, $f_3(\boldsymbol{b}'_3) = 1$. That is $f_i(\boldsymbol{b}'_j) = \delta_{ij}$.

(5) Elementary operations do not change the rank. B is obtained from I_3 by interchanging the first two rows, hence is invertible (Theorem 3.2), and Theorem 3.4 gives $\operatorname{rank}(BA) = \operatorname{rank}(A) = 2$. Directly:

$$BA = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

whose first row is twice the second while the remaining two rows are not proportional, so the rank is still 2.

Remark 4.1 (Not covered here — these belong to later lectures)**.**

- **Partitioned matrices and LU factorizations**: Lay §2.4, p. 123, §2.5, p. 132; Friedberg near §3.3–§3.4. Here elementary matrices only implement row operations; matrices are not yet split into a lower times an upper triangular factor.
- **Determinants**: Lecture 7 (Lay §3.1, p. 164; Friedberg §4.1). This lecture avoids them on purpose: similarity, rank and invertibility can all be discussed without them.
- **Eigenvalues and diagonalization**: Lecture 9 (Lay §5.2–§5.3; Friedberg §5.1–§5.2). Lecture 9 returns to similarity and asks which Q diagonalises $[T]_\beta$.
- **Inner products and adjoints**: Lecture 11 (Lay §6.1; Friedberg §6.1–§6.2). The geometric version of duality appears there.

5 Supplement: notions Homework 5 uses but the lecture did not cover

Remark 5.1 (Where these come from)**.** None of the following is part of the lecture's main thread, yet Homework 5 uses them:

- **Inclusion** and **quotient maps**: Problem 16 of the sheet (for V/W see the Lecture 2 notes).
- **Exactness** and **exact sequences**: Definitions 1 and 2 of the sheet.
- The **evaluation functional** $f_c(p) = p(c)$: Problem 14 of the sheet.
- **Change of basis between two spaces**: Theorem 1.4 covers operators only; the general form is Exercise 8 of Friedberg, §2.5 (printed p. 117–118).
- **Notation**: as in the Lecture 3 notes, the null space is $\mathcal{N}(T)$ and the range $\mathcal{R}(T)$, where the sheet and most texts write $\ker T$ and $\operatorname{Im} T$.

Definition 5.1 (Inclusion and quotient maps; homework sheet, Problem 16)**.** Let W be a subspace of a vector space V (for V/W see the Lecture 2 notes). The **inclusion map** is

$$S : W \to V, \qquad S(\boldsymbol{x}) = \boldsymbol{x},$$

and the **quotient map** is

$$T : V \to V/W, \qquad T(\boldsymbol{v}) = \boldsymbol{v} + W.$$

Both are linear: the first is the restriction of the identity of V to W, the second is well defined by the addition and scalar multiplication of cosets.

Definition 5.2 (Exactness; homework sheet, Definition 1)**.** Let $A \xrightarrow{S} B \xrightarrow{T} C$ be linear maps of vector spaces. The composite is **exact** at B when

$$\mathcal{N}(T) = \mathcal{R}(S),$$

that is, when $T(\boldsymbol{b}) = \boldsymbol{0}$ holds exactly for those $\boldsymbol{b}$ that are images of some $\boldsymbol{a} \in A$.

Definition 5.3 (Exact sequence; homework sheet, Definition 2)**.** A sequence

$$\cdots \to V_{i+1} \xrightarrow{T_{i+1}} V_i \xrightarrow{T_i} V_{i-1} \to \cdots$$

is **exact** when $\mathcal{N}(T_i) = \mathcal{R}(T_{i+1})$ for every i. The symbol 0 in such a sequence denotes both the zero vector space and the zero map ($\boldsymbol{x} \mapsto \boldsymbol{0}$).

Definition 5.4 (Evaluation functional; homework sheet, Problem 14)**.** Let $V = P_n(\mathbb{F})$ and $c \in \mathbb{F}$. Define

$$f_c : V \to \mathbb{F}, \qquad f_c(p) = p(c).$$

It is linear: $(p + q)(c) = p(c) + q(c)$ and $(ap)(c) = a\,p(c)$.

Remark 5.2 (Change of basis between two different spaces)**.** Theorem 1.4 treats an operator $T : V \to V$, where a single Q serves both ends. For a general $T : V \to W$ each end has its own bases, and

$$[T]^{\gamma'}_{\beta'} = P^{-1}[T]^{\gamma}_{\beta}Q,$$

with $Q = [I_V]^{\beta}_{\beta'}$ and $P = [I_W]^{\gamma}_{\gamma'}$ (Friedberg §2.5, Exercise 8, p. 117–118). The matrix then sits between two different invertible matrices rather than in the sandwich form $Q^{-1}AQ$, which comes back only when $V = W$ and the same Q is used at both ends.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. state the definitions of the coordinate vector and the coordinate mapping, and explain why the coordinate mapping is an isomorphism (uniqueness gives linearity, spanning gives surjectivity, independence gives injectivity);
2. write down the change of coordinate matrix $Q = [I_V]^{\beta}_{\beta'}$, say that its columns are the vectors of β' in β, and state $[\boldsymbol{v}]_{\beta} = Q[\boldsymbol{v}]_{\beta'}$ on the spot;
3. change the basis of a 2×2 operator with $Q^{-1}[T]_{\beta}Q$ and check the result against the images of the basis vectors;
4. define similar matrices, prove that similarity is an equivalence relation, and say that rank and invertibility are invariant under similarity;
5. give the three examples of linear functionals (integral, trace, coordinate) and explain why $\dim V^* = \dim V$;
6. compute the dual basis of a given basis by solving the δ_{ij} equations, and say precisely what "the two bases are dual to each other" means;
7. state $T^t(g) = g \circ T$ and $[T^t]^{\beta^*}_{\gamma^*} = ([T]^{\gamma}_{\beta})^{\top}$, and explain why "the matrix of the transpose is the transpose of the matrix";
8. state the double dual theorem and explain why ψ is natural while an isomorphism $V \cong V^*$ is not;
9. compute the rank of a matrix by counting pivots in a row reduction, or by counting a maximal independent set of columns, and pass between the two descriptions;
10. say that rank is preserved by multiplication by invertible matrices and by elementary operations, and state the Rank Theorem $\operatorname{rank}(A) + \dim \operatorname{Nul}(A) = n$;
11. use the rank inequalities (Theorem 3.10) to bound the rank of a composite or a product from above and below.

Exercises (printed pages). Lay: §4.4 p. 222 (Exercises 1–13: coordinate vectors and the coordinate mapping), §4.7 p. 242 (Exercises 1–9: change-of-coordinates matrices), §4.6 p. 236 (Exercises 1–23: rank, row space, the Rank Theorem), §2.2 p. 108 (Exercises 1–21: elementary matrices and inverses), §2.3 p. 114 (exercises on the Invertible Matrix Theorem); Friedberg: §2.5 p. 116 (Exercises 2, 3, 6, 8: change of basis and similarity), §2.6 p. 123 (Exercises 2–7: dual bases and the transpose), §3.1 p. 151 (Exercises 1–8: elementary matrices), §3.2 p. 165 (Exercises 1–15: rank, normal form, rank inequalities). Each section also has Practice Problems; do those first.

How to use the two texts together:

- **Compute and prove are divided cleanly.** How to find a change of coordinate matrix, how to count a rank: Lay (§4.4, §4.7, §4.6). Why the change of basis is $Q^{-1}AQ$, why the double dual is natural, why the rank does not depend on the representation: Friedberg (§2.5, §2.6, §3.2).
- The core chain of the lecture: *the coordinate mapping is an isomorphism → the change of coordinate matrix* $Q = [I_V]^{\beta}_{\beta'}$ *→ the operator changes basis as* $[T]_{\beta'} = Q^{-1}[T]_{\beta}Q$ *→ similarity*

is "the operator in different clothes" → the rank survives the change of clothes. The second thread is *duality*: *linear functionals → the dual basis → the transpose → the double dual is natural.*

- The single most important statement is Theorem 3.3: it welds "the rank of an operator" to "the rank of its matrix", so that the rank is at once a geometric quantity (the dimension of the image) and a computable one (the number of pivots), and Theorem 3.4 says it is also an **invariant**.